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Problems on Ages

SOLVED EXAMPLES

Ex. 1. The ratio of the ages of Tina and Rakesh is 9 : 10 respectively. Ten years ago, the ratio of their ages was 4 : 5 respectively. What is the present age of Rakesh? (Bank P.O., 2010)

Sol. Let Tina's age be $9x$ years. Then, Rakesh's age = $10x$ years.

$$\therefore \frac{9x - 10}{10x - 10} = \frac{4}{5} \Rightarrow 5(9x - 10) = 4(10x - 10) \Rightarrow 45x - 40x = 50 - 40 \Rightarrow 5x = 10 \Rightarrow x = 2.$$

$\therefore$ Present age of Rakesh = (10×2) years = 20 years.

Ex.2. Samir's age is one-fourth of his father's age and two-third of his sister Reema's age. What is the ratio of the ages of Samir, Reema and their father respectively? (Bank P.O., 2009)

Sol. Samir's age = $\frac{1}{4} \times$ Father's age = $\frac{2}{3} \times$ Reema's age = x years (say).

Then, Samir's age = x years, Reema's age = $\frac{3}{2}x$ years and Father's age = $4x$ years.

Ratio of the ages of Samir, Reema and father = $x : \frac{3}{2}x : 4x = 2 : 3 : 8$.

Ex.3. The age of father 10 years ago was thrice the age of his son. 10 years hence father's age will be twice that of his son. Find the ratio of their present ages. (L.I.C., 2003)

Sol. Let son's age 10 years ago be x years. Then, father's age 10 years ago = $3x$ years.

Son's age 10 years hence = $(x + 20)$ years.

Father's age 10 years hence = $(3x + 20)$ years.

$$\therefore 3x + 20 = 2(x + 20) \Rightarrow x = (40 - 20) = 20.$$

Ratio of father's age and son's age at present = $(3x + 10) : (x + 10)$
 $= (3 \times 20 + 10) : (20 + 10) = 70 : 30 = 7 : 3$.

Ex. 4. A man's present age is two-fifths of the age of his mother. After 8 years, he will be one-half of the age of his mother. How old is the mother at present? (M.A.T., 2009)

Sol. Let mother's age be x years. Then, man's age = $\frac{2x}{5}$ years.

$$\frac{2x}{5} + 8 = \frac{1}{2}(x + 8) \Rightarrow \frac{2x}{5} + 8 = \frac{1}{2}x + 4 \Rightarrow \frac{1}{2}x - \frac{2x}{5} = 4 \Rightarrow 5x - 4x = 40 \Rightarrow x = 40.$$

$\therefore$ Mother's age = 40 years.

Ex. 5. The ages of two persons differ by 16 years. If 6 years ago, the elder one be three times as old as the younger one, find their present ages. (L.I.C., 2009)

Sol. Let their ages be x years and $(x - 16)$ years. Then,

$$(x - 6) = 3 \{(x - 16) - 6\} \Rightarrow x - 6 = 3(x - 22) \Rightarrow 3x - x = 66 - 6 \Rightarrow 2x = 60 \Rightarrow x = 30.$$

So, their ages are 30 years and $(30 - 16) = 14$ years.

Ex. 6. The product of the ages of Ankit and Nikita is 240. If twice the age of Nikita is more than Ankit's age by 4 years, then find Nikita's age.

Sol. Let Ankit's age be x years. Then, Nikita's age = $\frac{240}{x}$ years. $2 \times \frac{240}{x} - x = 4 \Rightarrow \frac{480}{x} - x = 4 \Rightarrow 480 - x^2 = 4x$

$$\Rightarrow x^2 + 4x - 480 = 0$$

$$\Rightarrow x^2 + 24x - 20x - 480 = 0 \Rightarrow x(x + 24) - 20(x + 24) = 0$$